

CURRICULUM VITAE

NAME: Xi ZHANG

AFFILIATION: Massachusetts General Hospital/Harvard Medical School

EMAIL: xzhang84@mgh.harvard.edu

MOBILE: 7815188746

RESEARCH EXPERIENCES

- | | |
|-----------------|---|
| 09/2023-present | Research fellow <ul style="list-style-type: none">• Massachusetts General Hospital/Harvard Medical School |
| | Visiting scholar |
| 09/2020-09/2021 | <ul style="list-style-type: none">• Institute of Biomedical Engineering, Shenzhen Bay Laboratory, China• Supervisor: Dr. Qiyu Peng and Prof. Qiushi Ren |
| | Visiting scholar |
| 09/2017-03/2018 | <ul style="list-style-type: none">• Molecular Biophysics and Integrated Bioimaging, Lawrence Berkeley National Laboratory, USA• Supervisor: Dr. Qiyu Peng and Prof. Thomas F. Budinger |

EDUCATION

- | | |
|-----------------|--|
| | Ph.D. in Mechanical electronics |
| 09/2016-12/2022 | <ul style="list-style-type: none">• Supervisor: Prof. Jianfeng Xu• Huazhong University of Science and Technology, China |
| | B.S. in Mechanical engineering |
| 09/2012-06/2016 | <ul style="list-style-type: none">• Huazhong University of Science and Technology, China |

PROJECT EXPERIENCE

[1] Development of a Polyhedral Structure High-performance Molecular Imaging Instrument for Brain Science Research. 01/2017-12/2021, National Natural Science Foundation of China (Major Program).

[2] The Research of Core Components and Critical Techniques for PET/MRI Dual-modality Imaging. 01/2016-12/2019, National Natural Science Foundation-Guangdong Joint Funds of China.

[3] **Exploring the Benefits of Time-of-Flight PET**, 04/2013-03/2019, National Institute of Health (NIH) R01 project.

SKILLS

- Coding language: C++, Python, Verilog HDL
- Familiar with Software: MATLAB, Quartus (Verilog), GEANT 4, Altium Designer, Cadence Allegro, Solidworks, Auto CAD
- Proficient in PET detector design: TOF and DOI

HONORS AND AWARDS

- [1] 2021, Merit postgraduate of HUST
- [2] 2017, National scholarship
- [3] 2016, Outstanding Graduates of HUST
- [4] 2014, First prize of "Challenge Cup" National College Student Business Plan Competition
- [5] 2014, National Encouragement Scholarship
- [6] 2013, Merit Student of HUST

PUBLICATIONS

- [1] **Zhang X**, Sitek A, Blackberg L, Kupinski M, Furenlid L, Sabet H. Development and Performance Evaluation of a Laser Processed CsI:Tl Detector with Converging Pixels. In: 2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD). 2025:1-1. doi:10.1109/NSS/MIC/RTSD57106.2025.11286546
- [2] **Zhang X**, Feng Y, Ottensmeyer MP, Sabet H. Modular and Robust PET Platform for Evaluating Detector Configurations. In: 2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD). 2025:1-1. doi:10.1109/NSS/MIC/RTSD57106.2025.11287227
- [3] Hashemi A, **Zhang X**, Kouame D, Sabet H. Enhancing Detector Event Positioning Data Using L1 Regularization. In: 2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD). 2025:1-2. doi:10.1109/NSS/MIC/RTSD57106.2025.11287406

- [4] **Zhang X**, Yu X, Cheng H, et al. Enhancing gamma-ray detection: Processing grooved microstructures on LYSO crystal with femtosecond laser. *Med Phys*. 2025;52(4):2644-2660. doi:10.1002/mp.17665
- [5] Yu X, **Zhang X**, Zeng J, et al. Depth of Interaction in PET Detector Design: Performance Optimization With Light-Sharing Window. *IEEE Trans Instrum Meas*. 2025;74:1-10. doi:10.1109/TIM.2024.3522335
- [6] Cheng H, Bai L, **Zhang X**, et al. ⁶⁸Ga labeled Olmutinib: Design, synthesis, and evaluation of a novel PET EGFR probe. *Bioorganic Chem*. 2024;153:107987. doi:10.1016/j.bioorg.2024.107987
- [7] Cheng H, Liu R, Fang S, Li Z, Zhang D, **Zhang X** et al. Synthesis of easily-modified and useful dibenzo-[b,d]azepines by palladium(ii)-catalyzed cyclization/addition with a green solvent. Published online March 26, 2024. doi:10.1039/D3CC06321F
- [8] **Zhang X**, Yu X, Zhang H, et al. Development and Evaluation of 0.35-mm-Pitch PET Detectors With Different Reflector Arrangements. *IEEE Trans Radiat Plasma Med Sci*. 2023;7(8):802-809. doi:10.1109/TRPMS.2023.3307128
- [9] Yu X, **Zhang X**, Zhang H, et al. Requirements of Scintillation Crystals with the Development of PET Scanners. *Crystals*. 2022;12(9). doi:10.3390/cryst12091302
- [10] **Zhang X**, Yu X, Zhu Z, et al. Development and Evaluation of a Dual-Layer-Offset PET Detector Constructed with Different Reflectors. *Crystals*. 2022;12(1). doi:10.3390/cryst12010093
- [11] Xie S, Zhu Z, **Zhang X**, et al. Optical Simulation and Experimental Assessment with Time–Walk Correction of TOF–PET Detectors with Multi-Ended Readouts. *Sensors*. 2021;21(14). doi:10.3390/s21144681
- [12] **Zhang X**, Yu H, Xie Q, et al. Design study of a PET detector with 0.5 mm crystal pitch for high-resolution preclinical imaging. *Phys Med Biol*. 2021;66(13):135013. doi:10.1088/1361-6560/ac0b82
- [13] **Zhang X**, Xie Q, Xie S, Yu X, Xu J, Peng Q. A Novel Portable Gamma Radiation Sensor Based on a Monolithic Lutetium-Yttrium Oxyorthosilicate Ring. *Sensors*. 2021;21(10). doi:10.3390/s21103376
- [14] **Zhang X**, Ye B, Yu H, et al. Depth of Interaction Measurements Based on Rectangular Light Sharing Window Technology and Nine-Crystals-to-One-SiPM Coupling Method. *IEEE Trans Radiat Plasma Med Sci*. 2021;5(3):306-314.

- [15] **Zhang X**, Ye B, Xie S, et al. Preliminary Optimized Design of a High-resolution PET Detector with a 0.5 mm Crystal Size. In: 2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). 2020:1-2. doi:10.1109/NSS/MIC42677.2020.9507797
- [16] Yang J, Xie Q, Xie Y, **Zhang X**, et al. A Simulation Study on Two Readout Methods of a Monolithic Scintillator Ring PET. In: 2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). 2020:1-3. doi:10.1109/NSS/MIC42677.2020.9508071
- [17] Xie Q, **Zhang X**, Xie Y, Xie S, Peng Q, Xu J. A Novel Portable Radiation Detector Based on Monolithic LYSO Ring: A Simulation Study. In: 2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). 2020:1-4. doi:10.1109/NSS/MIC42677.2020.9507999
- [18] Jiao W, **Zhang X**, Xie S, Ying G, Xu J, Peng Q. A Novel Method for Processing Photonic Crystals: Femtosecond Laser. In: 2020 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). 2020:1-2. doi:10.1109/NSS/MIC42677.2020.9507800
- [19] Xie S, **Zhang X**, Zhang Y, et al. Evaluation of Various Scintillator Materials in Radiation Detector Design for Positron Emission Tomography (PET). Crystals. 2020;10(10). doi:10.3390/cryst10100869
- [20] Xie S, **Zhang X**, Huang Q, Gong Z, Xu J, Peng Q. Methods to Compensate the Time Walk Errors in Timing Measurements for PET Detectors. IEEE Trans Radiat Plasma Med Sci. 2020;4(5):555-562. doi:10.1109/TRPMS.2020.2981388
- [21] Song Z, Zhao Z, Yu H, Yang J, **Zhang X**, et al. An 8.8 ps RMS Resolution Time-To-Digital Converter Implemented in a 60 nm FPGA with Real-Time Temperature Correction. Sensors. 2020;20(8). doi:10.3390/s20082172
- [22] **Zhang X**, Ye B, Xie S, et al. A Novel Preclinical PET Scanner Constructed with Stacks of Sliced Monolithic Scintillator Rings. In: 2019 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). 2019:1-2. doi:10.1109/NSS/MIC42101.2019.9059816
- [23] Xie Q, Ye B, Xie S, **Zhang X**, Xu J, Peng Q. Design, fabrication and evaluation of anti-reflection films coated on LYSO scintillators. In: 2019 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC). 2019:1-2. doi:10.1109/NSS/MIC42101.2019.9059696
- [24] Zhao Z, Xie S, **Zhang X**, et al. An Advanced 100-Channel Readout System for Nuclear Imaging. IEEE Trans Instrum Meas. 2019;68(9):3200-3210. doi:10.1109/TIM.2018.2877952
- [25] Xie S, Xu J, Yang M, Ying G, **Zhang X**, et al. Methods to Improve Light Transport Efficiency in LYSO Crystals Based on Characteristics of Optical Reflectance. IEEE Trans Nucl Sci.

2019;66(9):2100-2106. doi:10.1109/TNS.2019.2931929

[26] Xu J, Xie S, **Zhang X**, et al. A preclinical PET detector constructed with a monolithic scintillator ring. *Phys Med Biol*. 2019;64(15):155009. doi:10.1088/1361-6560/ab2ca4

[27] Xie S*, **Zhang X***, Peng H, et al. PET detectors with 127 ps CTR for the Tachyon-II time-of-flight PET scanner. *Nucl Instrum Methods Phys Res Sect Accel Spectrometers Detect Assoc Equip*. 2019;933:48-55. doi:10.1016/j.nima.2019.03.083

[28] **Zhang X**, Xie S, Yang J, et al. A depth encoding PET detector using four-crystals-to-one-SiPM coupling and light-sharing window method. *Med Phys*. 2019;46(8):3385-3398. doi:10.1002/mp.13603

[29] Ye B, **Zhang X**, Xie Y, et al. Experimental studies of the decoding performances of a semi-monolithic scintillator detector. In: 2018 IEEE Nuclear Science Symposium and Medical Imaging Conference Proceedings (NSS/MIC). 2018:1-3. doi:10.1109/NSSMIC.2018.8824487

[30] Yang J, Xie Y, Xie S, **Zhang X**, et al. Experimental studies of the performance of different methods in the inter-crystal Compton scatter correction on one-to-one coupled PET detectors. In: 2018 IEEE Nuclear Science Symposium and Medical Imaging Conference Proceedings (NSS/MIC). 2018:1-3. doi:10.1109/NSSMIC.2018.8824396